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UNIT-4 : SENSORS

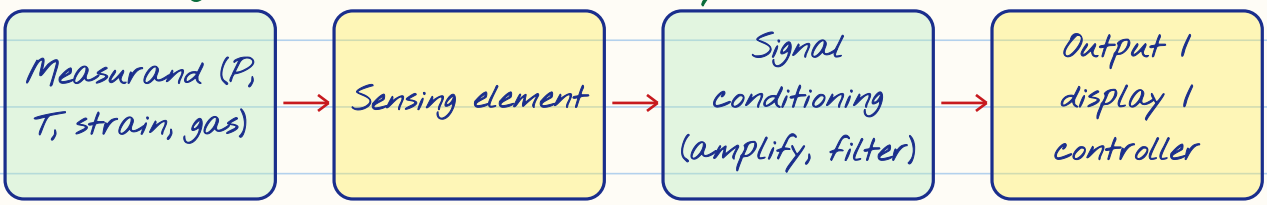
IMPORTANT: Syllabus: Introduction, Conductometric sensors, Capacitive sensors, Piezoelectric sensors, Magnetostrictive sensors, Piezoresistive sensors, Optical sensors, Resonant sensors, Semiconductor-based sensors, Acoustic sensors, Polymeric sensors, Carbon nanotube sensors.

4.1 Introduction to Sensors

Definition: A sensor is a device that detects a physical, chemical or biological quantity (measurand) and converts it into a measurable signal, usually electrical (voltage, current, resistance, capacitance, frequency).

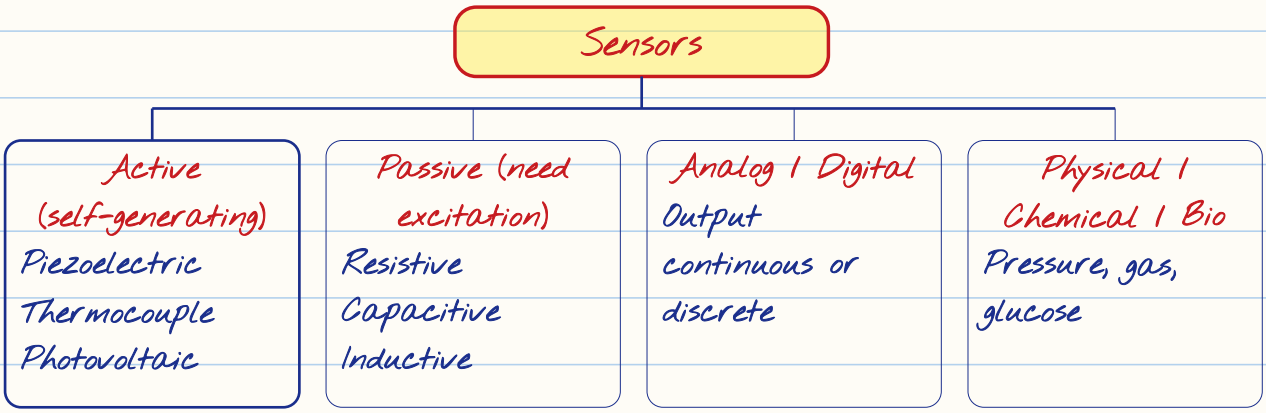
- A transducer converts one form of energy into another; a sensor is the input transducer, an actuator the output transducer.

Block diagram of a measurement system



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Classification of sensors



Static characteristics (performance parameters)

- **Range and span** : minimum - maximum measurable values.
- **Sensitivity** = change in output / change in input (slope of calibration curve).
- **Linearity** : closeness of calibration curve to a straight line.
- **Resolution** : smallest detectable change. **Accuracy** and **precision** (repeatability) .
- **Hysteresis** : output difference for increasing and decreasing input.
- **Response time** , rise time, **bandwidth** (dynamic).
- **Selectivity** (cross-sensitivity), **drift** , **offset** , noise, stability, **detection limit** .

IMPORTANT: Smart sensors = sensor + microprocessor + communication interface on the same chip (self-calibration, compensation, digital output, IoT ready).

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4.2 Conductometric Sensors

Definition: Conductometric (chemiresistive) sensors measure the change in electrical conductivity (or resistance) of a sensing material when it interacts with the analyte (gas, vapour, ion, humidity, temperature).

$$G = 1 / R = \sigma \times A / L$$ conductance

Principle

- The target gas adsorbs on the surface of the sensing layer and exchanges electrons, changing the carrier concentration → conductance changes.
- n-type SnO_2 in air: oxygen adsorbs, traps electrons → high resistance. Reducing gas (CO, H_2, CH_4, ethanol) reacts with oxygen species and releases electrons → resistance falls. Oxidising gas (NO_2) → resistance rises.

Construction

- Ceramic substrate with interdigitated electrodes (IDE) (Pt, Au) + sensing film (SnO_2, ZnO, WO_3, TiO_2, In_2O_3, conducting polymer, CNT) + micro-heater (Pt) to keep 200–400 C.

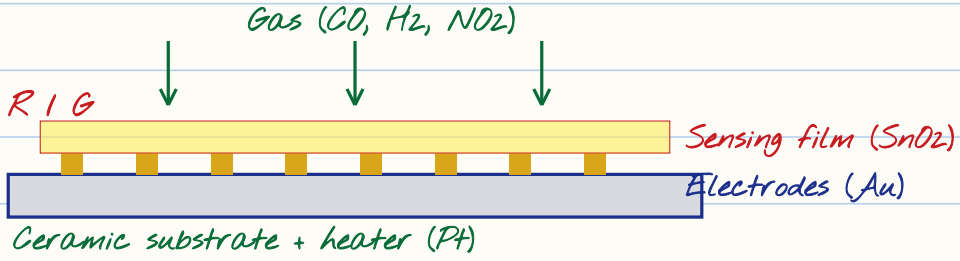


Fig: Metal-oxide conductometric gas sensor

- Limitations: poor selectivity, high power for heater, drift, humidity and temperature effects, poisoning.

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4.3 Capacitive Sensors

Definition: A capacitive sensor measures a change in capacitance caused by a change in plate distance, overlap area or dielectric of a capacitor.

$$C = eps0 \times eps_r \times A / d$$ parallel plate

- $eps0 = 8.85 \times 10^{-12}$ F/m, eps_r = relative permittivity, A = overlap area, d = gap.

Three sensing mechanisms

Variable	Change in	Use
Gap (d)	Distance	Pressure, displacement, accelerometer, microphone (non-linear)
Area (A)	Overlap	Angular / linear position, liquid level (linear)
Dielectric (eps_r)	Material between plates	Humidity, level, moisture, touch

- Differential configuration (two capacitors, one increases, one decreases) gives linear output, double sensitivity and reduces temperature errors.

Examples

- MEMS capacitive pressure sensor : a thin silicon / polysilicon diaphragm deflects under pressure, reducing the gap to the fixed electrode → C increases.
- MEMS capacitive accelerometer : proof mass on springs between fixed plates (eg. ADXL series); acceleration moves the mass → C changes; used in air bags and phones.
- Capacitive humidity sensor : polymer (polyimide) film absorbs water → eps_r (water 80) increases → C increases.
- Limitations: stray / parasitic capacitance , small capacitance

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(pF-fF) → electronics must be close, non-linear for gap variation, humidity and dust effects.

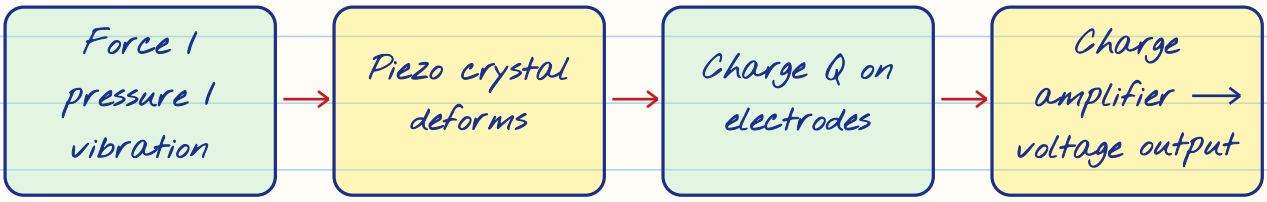
4.4 Piezoelectric Sensors

Definition: Piezoelectric sensors use the direct piezoelectric effect : mechanical stress / strain / force / acceleration applied on a piezo material generates electric charge proportional to the applied quantity.

$Q = d \times F$ and $V = Q / C$ charge and voltage

- Quartz (stable), PZT (high sensitivity), PVDF (flexible film), ZnO and AlN thin film (MEMS), lithium niobate, PMN-PT single crystal.

Working



- High internal impedance → charge amplifier or voltage amplifier with high input impedance (FET); IEPE / ICP sensors have built-in electronics.

Characteristics

- Dynamic sensor - responds to changing (AC) quantities only; not suitable for true static measurement because charge leaks away (low-frequency limit ~ 0.1-1 Hz).
- Wide frequency range (up to 100 KHz+), self-generating (no power), high stiffness, rugged, small size, wide temperature range (quartz, up to 500 C).

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Applications

- Accelerometers (vibration, crash test), pressure sensors (engine knock, combustion), force and load cells, ultrasonic transducers (NDT, medical), microphones, hydrophones, vibration and structural health monitoring, flow meters, touch pads, energy harvesting, piezo igniters, gyroscopes (tuning fork).

4.5 Magnetostrictive Sensors

Definition: Magnetostrictive sensors work on the Villari (inverse magnetostriction) effect and the magnetoelastic coupling: stress changes the magnetisation (permeability) of a ferromagnetic material, which is detected by a coil or a Hall / GMR element. Or they use the Wiedemann / Matteucci effects.

- Magnetoelastic force / torque sensor: stress in a Ni / Fe-Ni shaft changes its permeability → detected by pick-up coils (used for non-contact torque sensing in cars, steering).
- Magnetostrictive linear position sensor (Temposonics, Wiedemann effect):

Current pulse in waveguide wire → Interacts with magnet field of position magnet → Torsional (strain) pulse by Wiedemann effect → Pulse travels to the end, detected by coil → time of flight = position

  - Position $x = v \times t$ where $v = \sim 2800$ m/s. Resolution of microns; long strokes up to several metres; non-contact, wear free.
- Magnetostrictive resonant sensors (magnetoelastic ribbon, Metglas): strip resonates when excited by AC magnetic field;

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resonant frequency changes with mass load, temperature, pressure, viscosity, pH (with a coating) → wireless, passive sensing.

- Magnetostrictive microphone / pick up and magnetostrictive delay line.
- Limitations: needs magnetic field circuit, temperature-dependent, eddy current losses at high frequency, shielding.

4.6 Piezoresistive Sensors

Definition: Piezoresistivity is the change in electrical resistivity of a material when mechanical strain is applied. It is much larger in semiconductors (Si) than in metal strain gauges.

$$\frac{\Delta R}{R} = GF \times \text{strain}$$ GF = gauge factor

- Gauge factor GF: metal foil ~ 2; silicon ~ 100–200 (p-type crystalline), polysilicon 20–40.

Construction

- Metallic strain gauge: constantan / Karma foil grid bonded on the structure.
- Silicon piezoresistor: boron-doped p-type resistors diffused / implanted on a silicon diaphragm (MEMS pressure sensor) or on a cantilever beam.

Wheatstone bridge

- Four piezoresistors connected in a Wheatstone bridge (two in tension, two in compression) → output voltage proportional to pressure, temperature compensated, 4x sensitivity of single gauge.

$$V_{out} = V_s \times \left(\frac{\Delta R}{R}\right)$$ full bridge, ideal

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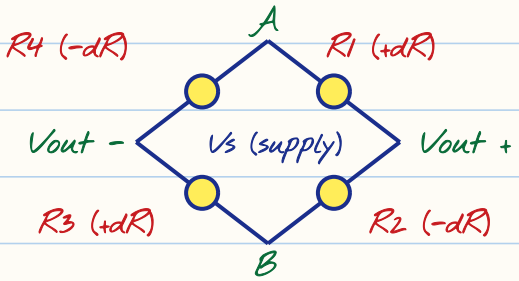


Fig: Full Wheatstone bridge with four piezoresistors

Characteristics

- High sensitivity, good linearity (< 0.1%), small size, mass production with IC technology, can integrate amplifier and temperature compensation on chip.
- Drawbacks: strong temperature dependence (TCR and TCGF), self-heating, requires compensation and calibration, fragile at high strain, offset drift.

Applications: Pressure sensors (automotive MAP, tyre, blood pressure), accelerometers, load cells, force and torque transducers, flow sensors, tactile sensors, structural monitoring (strain gauges), microphones, AFM cantilevers.

4.7 Optical Sensors

Definition: Optical sensors convert a measurand into a change in a property of light (intensity, wavelength, phase, polarisation, time) and detect it with a photodetector. Fibre-optic sensors (FOS) are the most important for smart structures.

Types of fibre optic sensors

- Intensity-based : bending loss (microbend), reflective (displacement), evanescent (gas / chemical). Simple and low cost, light source fluctuations disturb it.
- Phase-based (interferometric) : Mach-Zehnder, Michelson, Fabry-Perot (EFPI), Sagnac (fibre gyroscope). Extremely high sensitivity (nanometre); needs coherent source.

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- Wavelength-based - Fibre Bragg Grating (FBG) : periodic refractive-index modulation written in the core by UV light.

$$\lambda_B = 2\, n_{eff} \times \Lambda$$ Bragg wavelength

  - n_{eff} = effective refractive index, Λ = grating period. Strain or temperature changes Λ and n_{eff} → shift in reflected wavelength (strain ~ 1.2 pm / micro-strain; temperature ~ 10 pm / C at 1550 nm).

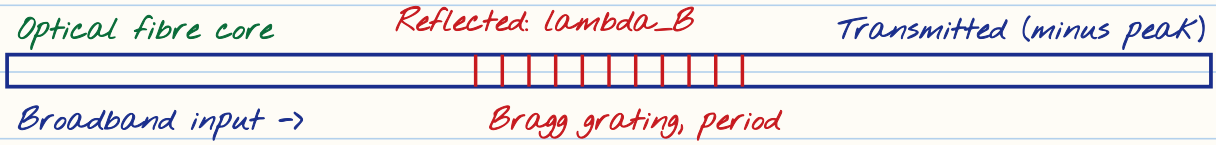


Fig: Fibre Bragg grating sensor

- Immune to EMI , light weight, thin (125 micron - can be embedded in composites), multiplexing (many FBGs on one fibre), intrinsically safe (no spark), high bandwidth, corrosion resistant, remote sensing, wide temperature range.
- Costly interrogators (spectrometer), temperature-strain cross-sensitivity, fragile at the ingress / egress, complex splicing.

Applications: Structural health monitoring (bridges, dams, aircraft wings, wind blades, pipelines, tunnels), temperature profiling, medical catheters, oil wells, fibre gyroscope, chemical / bio sensing, pH, smoke detectors, photo-electric sensors, proximity.

4.8 Resonant Sensors

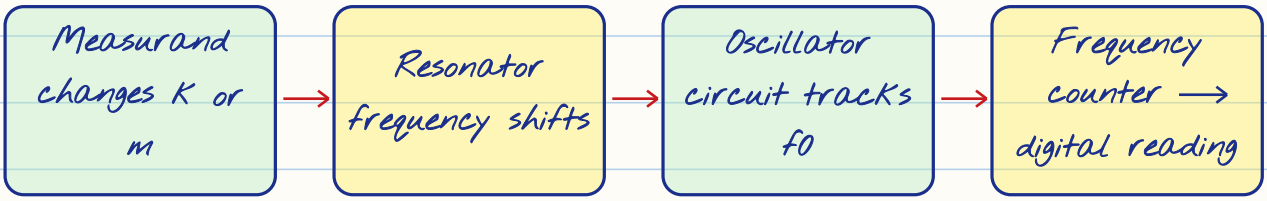
Definition: Resonant sensors consist of a mechanical resonator (beam, plate, tuning fork, membrane) whose natural (resonant) frequency changes with the measurand . The output is a frequency signal , which is quasi-digital.

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$$f0 = (1 / 2\pi) \times \sqrt{(K / m)}$$ mass-spring resonator

Principle

- Measurand (mass added, stress / tension, pressure, temperature, viscosity) changes either the mass m or the stiffness K of the resonator → change in f0.
- Excitation and detection: electrostatic, piezoelectric, electrothermal, magnetic or optical ; closed in a feedback loop with an amplifier (self-oscillating).



Examples

- Quartz crystal microbalance (QCM) - mass sensing; Sauerbrey equation : $\Delta f = -C_f \times \Delta m$ (frequency decreases when mass is added).
- Silicon resonant pressure sensor : diaphragm stress changes beam stiffness → frequency (Yokogawa DPharp, Druck).
- MEMS cantilever mass / biosensor (femtogram), vibrating wire strain gauge (civil), tuning-fork gyroscope , vibrating tube density meter , resonant accelerometer.
- Advantages: high accuracy / stability, direct digital (frequency) output , immune to amplitude noise, high resolution.
- Limitations: needs vacuum packaging for high Q, complex oscillator, temperature dependence, damping in liquids, non-linearity at large amplitude.

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4.9 Semiconductor-based Sensors

- Sensors made in / on silicon and other semiconductors, using semiconductor effects with IC technology.

Sensor	Principle	Application
Temperature (diode, transistor, band-gap)	VBE of a PN junction falls ~ 2 mV / C	IC temperature sensor (LM35)
Hall sensor	Lorentz force on carriers → Hall voltage VH = (I B) / (n q t)	Position, current, speed, compass
Photodiode / phototransistor / solar cell	Photon generates electron-hole pairs	Light, camera, optical comm
ISFET (ion-sensitive FET)	Surface potential depends on pH / ion concentration	pH, biosensing, DNA
Piezoresistive / capacitive MEMS	Si diaphragm, cantilever	Pressure, acceleration

- ISFET: gate oxide (SiO_2 / Si_3N_4 / Ta_2O_5) in contact with electrolyte; pH changes the surface charge → shifts threshold voltage VT ~ 50–59 mV / pH.
- Limitations: temperature sensitivity, limited high-temperature use (Si up to 150 C; use SiC / GaN for harsh), cross-sensitivity.

4.10 Acoustic Sensors

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Definition: Acoustic wave sensors use mechanical (acoustic) waves propagating in or on a piezoelectric substrate; any perturbation of the path (mass, viscosity, temperature, stress) changes the velocity or frequency of the wave. Excitation and detection are done with interdigital transducers (IDT) by piezoelectric effect.

Types

- BAW - Bulk Acoustic Wave : wave travels through the bulk; QCM (quartz crystal microbalance), thickness-shear mode (AT-cut quartz); FBAR (film bulk acoustic resonator) 1-10 GHz.
- SAW - Surface Acoustic Wave : Rayleigh wave confined to the surface (within one wavelength); substrate $LiNbO_3$, quartz, $LiTaO_3$, ZnO / AlN films; 100 MHz - 1 GHz.
  - SAW delay line or resonator sensor → frequency / phase shift.

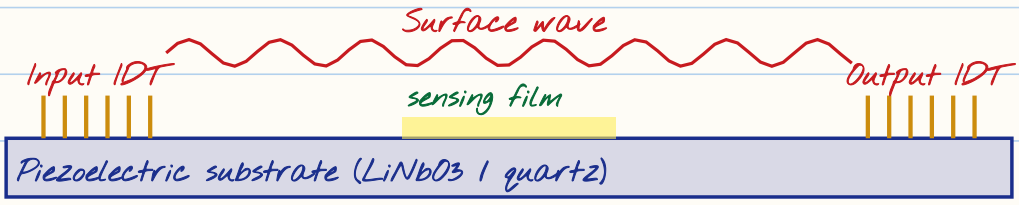


Fig: SAW delay line sensor

Sensing mechanism

- Mass loading : coated film (polymer) absorbs gas → mass increases → frequency drops (gas, vapour, biosensor).
- Temperature, pressure, torque, strain, humidity, viscosity measured by change in velocity.
- Advantages: very high sensitivity, small, rugged, wireless / passive (RFID type) operation, frequency output, low cost batch production.

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- Limitations: needs a coating for selectivity, aging of polymer film, damping in liquids (except Love / SH-modes), temperature drift.

Applications: Gas and vapour sensors (explosives, nerve agents), biosensors (antibody-antigen), wireless temperature in rotating parts, tyre pressure, touch screens (acoustic), RF filters in mobiles, ultrasonic NDT, microphones, sonar.

4.11 Polymeric Sensors

Definition: Polymer sensors use polymers (conducting, piezoelectric, dielectric or ionic) as the sensing layer ; the polymer interacts with the measurand and changes its resistance, capacitance, volume, optical property or charge output .

Types

- Conducting polymers (polyaniline PANI, polypyrrole PPy, polythiophene, PEDOT:PSS): gas (NH_3, NO_2), vapour, pH and bio-sensing - doping / de-doping changes conductivity; operate at room temperature .
- Piezoelectric polymers - PVDF, P(VDF-TrFE) : flexible force, pressure, strain, vibration, tactile and acoustic sensors.
- Polymer composite sensors : carbon black / CNT / graphene in insulating polymer → piezoresistive or chemiresistive (swells with vapour, conducting path breaks). Also pressure-sensitive rubber (Velostat) , e-skin.
- Molecularly imprinted polymers (MIP) : selective cavities for target molecules (biosensing).
- Light weight, flexible, low cost , easy processing (printing, spin coating), room temperature operation, large area, biocompatible, tunable chemistry.

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- Poor long-term stability, aging, drift, humidity and temperature effects, low selectivity, slow response, low operating temperature.

4.12 Carbon Nanotube (CNT) Sensors

Definition: Carbon nanotubes are rolled-up graphene sheets (cylinders of carbon atoms), diameter 1-100 nm, length microns. SWCNT (single-wall) and MWCNT (multi-wall); depending on chirality they can be metallic or semiconducting.

Properties that favour sensing

- Huge surface-to-volume ratio, every atom is on the surface → very sensitive to adsorption of molecules.
- Excellent electrical conductivity (current density 10^9 A/cm2), high mechanical strength (Young's modulus ~ 1 TPa, tensile 50-100 GPa), high thermal conductivity, chemical stability, and strain-dependent band-gap.

Types of CNT sensors

- Chemiresistor gas sensor : CNT network between electrodes - gas molecules (NO_2, NH_3) transfer charge → resistance changes; works at room temperature, ppb detection.
- CNT-FET (field effect transistor) sensor : a semiconducting CNT bridging source and drain; gas or biomolecule at the surface changes the gate threshold → big current change; used for DNA, protein, glucose detection.
- Strain / pressure sensors : CNT / polymer composite or aligned CNT film - piezoresistive; gauge factor up to 10-100; stretchable and wearable.

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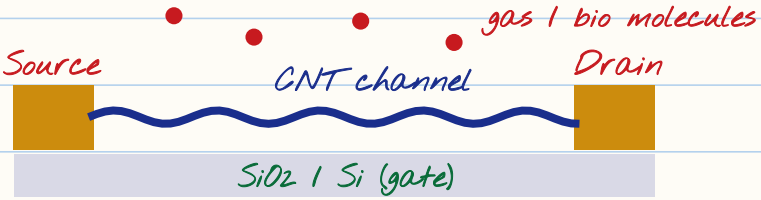


Fig: CNT field-effect transistor sensor

- Ultra-high sensitivity, fast response, low power, room temperature, miniature (nano), flexible integration, wide range of analytes.
- Difficult selective / chirality-controlled growth and purification, device-to-device variation, poor selectivity (needs functionalisation), cost, health and environmental concerns, drift due to humidity.

4.13 Quick Revision / Important Questions

- 1. Define sensor and explain static characteristics of a sensor.
- 2. Explain the working of a conductometric gas sensor.
- 3. Discuss capacitive sensing principles with a MEMS accelerometer.
- 4. Piezoelectric sensors: principle, equivalent circuit, signal conditioning, applications.
- 5. Explain piezoresistive pressure sensor with Wheatstone bridge.
- 6. Fibre Bragg grating sensor: principle and applications in SHM.
- 7. Differentiate BAW and SAW sensors. Applications of acoustic sensors.
- 8. Write short notes on polymer sensors and CNT sensors.

IMPORTANT: Exam tip: always write the principle equation (C = eps A / d, Q = d F, lambda_B = 2 n Lambda, delta R / R = GF x strain) and one application for each sensor.